Algebra 1 Name Answer Key________
Unit 7 Date____________________
Lesson 4 Practice Period___________________

For questions 1-5, factor each quadratic expression using the method of your choice.

1. $5b^2 + 46b + 48$

$5b^2 + 40b + 6b + 48$
$5b\,(b + 8) + 6(b + 8)$
$(5b + 6)\,(b + 8)$

2. $4a^2 + 27a + 18$

$4a^2 + 24a + 3a + 18$
$4a\,(a + 6) + 3(a + 6)$
$(4a + 3)\,(a + 6)$

3. $2x^2 + 3x - 27$

$2x^2 + 9x - 6x - 27$
$x\,(2x + 9) - 3\,(2x + 9)$
$(x - 3)\,(2x + 9)$

4. $3k^2 + 5k - 28$

$3k^2 + 12k - 7k - 28$
$3k\,(k + 4) - 7\,(k + 4)$
$(3k - 7)\;(k + 4)$

5. $5z^2 - 28z - 96$

$5z^2 - 40z + 12z - 96$
$5z(z - 8) + 12\,(z - 8)$
$(5z + 12)(z - 8)$

For questions 6-7, consider the trinomial $4x^2 - 41x + 72$.

6. Complete the area model.

	$4x$	-9
x	$4x^2$	$-9x$
-8	$-32x$	72

7. What are the two binomial factors for $4x^2 - 41x + 72$?

$(4x - 9)(x - 8)$

For questions 8-9, consider the trinomial $2m^2 - 9m - 110$.

8. Complete the area model.

	$2m$	$+11$
m	$2m^2$	$11m$
-10	$-20m$	-110

9. What are the two binomial factors for $2m^2 - 9m - 110$?

$(2m + 11)(m - 10)$

10. Factor $3x^2 - 4x - 32$ using factoring by grouping.

$3x^2 - 12x + 8x - 32$
$3x\,(x - 4) + 8\,(x - 4)$
$(3x + 8)(x - 4)$